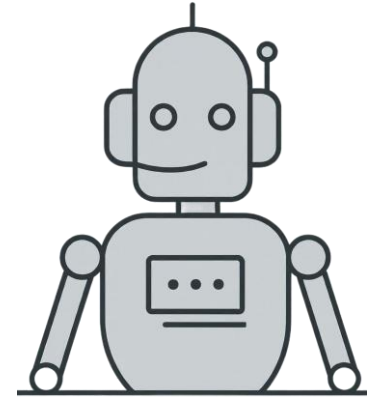


Building No Code Gen AI Workflows

Week 10: Evaluating Agentic AI Workflows – (recap)



Mike Lively

<https://github.com/eaglelandsonce/week10recapnocode>



Mike Lively

Founder QuantumAI
Trainer, IT Evangelist,
Developer
@mikelively-quantumai



18 Azure Cloud Certs



About Me

Father of Nine, USAF
Working on a PhD in GenAI
Teaching AI for Johns Hopkins & GK
Avid Hacker & Prompt Engineer
40 years programming, Keynote
Flautist
YMCA Gym Rat

PhD "ABD"





ICE Breaker

Prompt 1: AI given what you know of me what are the tools I would need if I were an agent to accomplish my daily tasks.

Prompt 2: Draw an image of my tools in action!

Does this match your weakness, or is it off base?

Agenda

Learning Objectives

Copilot Help Desk

What is wrong?

Why this error cascades?.

Shopease Hands-On Step-by-Step

Slides and Prompts will be provided

Mobile Push (Walk Through – How you Do It!)

Adding mobile push

Adding mobile push to the simple story maker

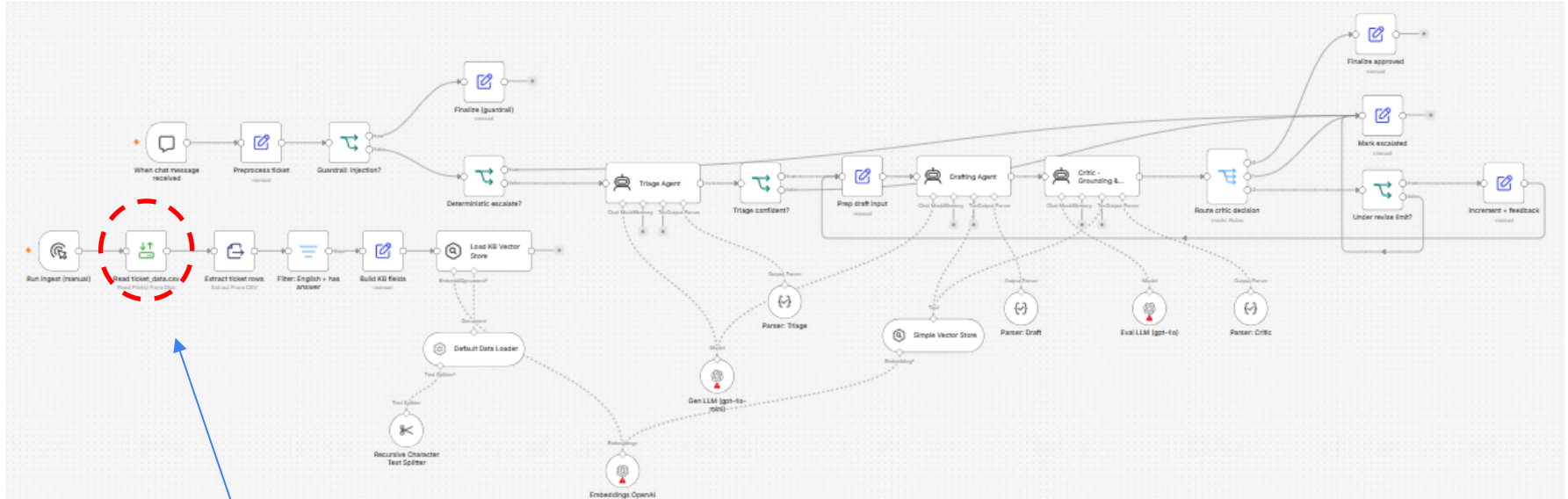
Session Summary

<https://github.com/eaglelandsonce/week10recapnocode>

Case Study

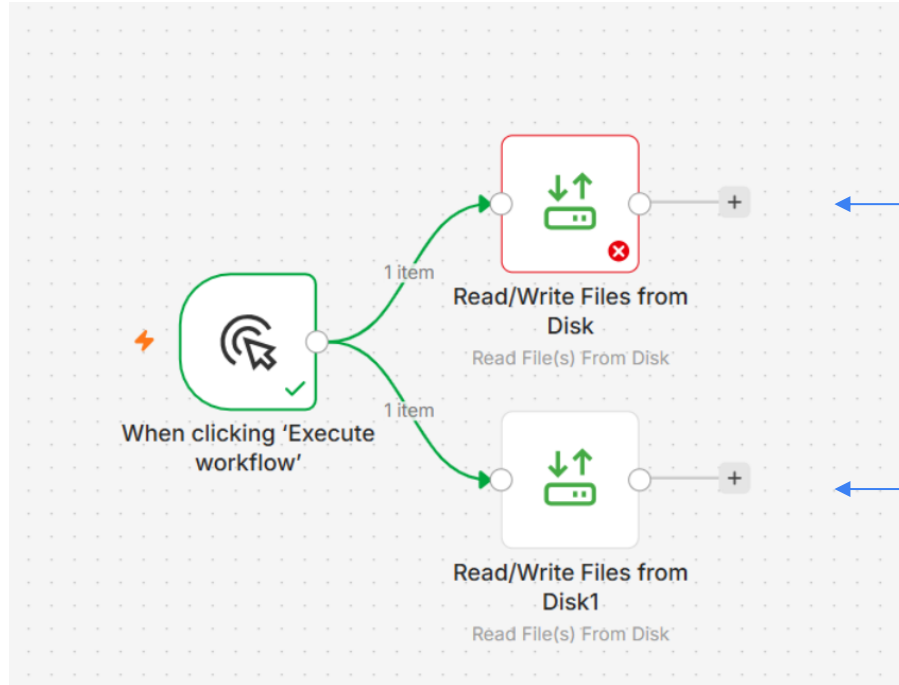
Helpdesk Copilot What's Wrong

Copilot Help Desk Workflow



Where the Issue is!

Copilot Help Desk Workflow



From lab 10

Most likely came with
the software update!

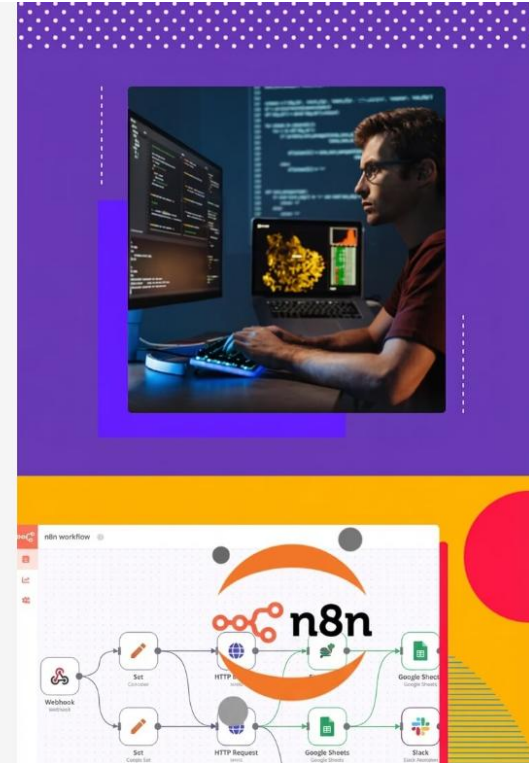
From lab 9



Welcome to the n8n Lab

- ✓ **Visual Workflow Building**
Design, test, and orchestrate node-based automations interactively using the n8n lab
- ✓ **Save and Resume**
Your workflows are saved and can be resumed or iterated on later
- ✓ **Export**
Export workflows for deployment or future reuse

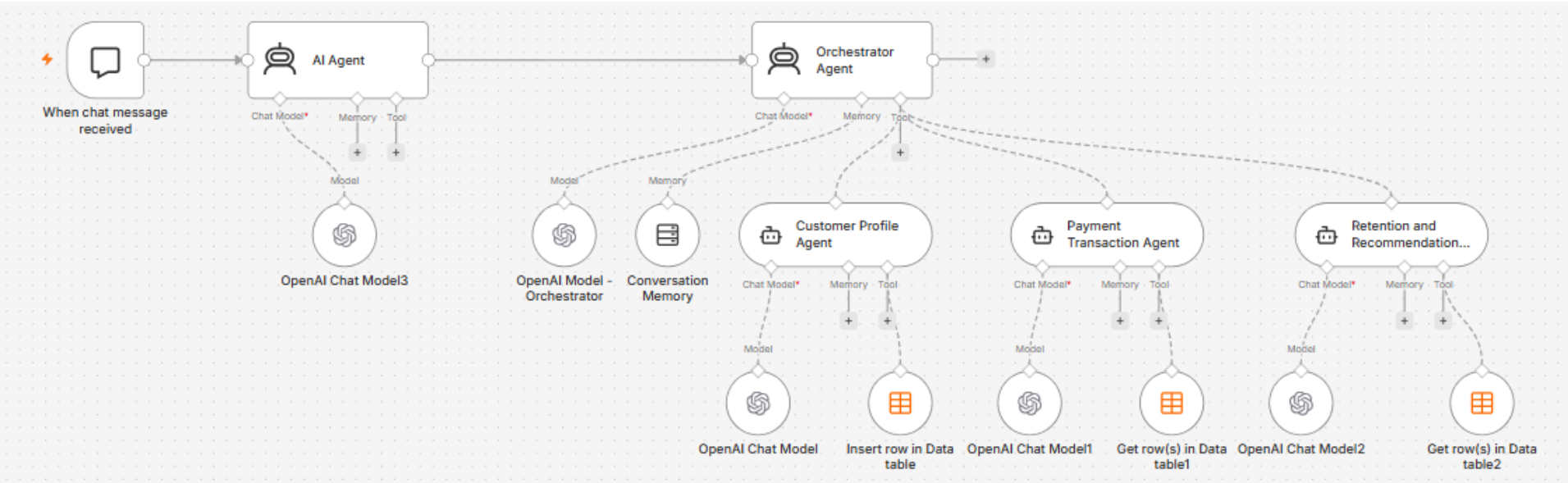
RESUME N8N LAB



https://olympus.mygreatlearning.com/gl_coding_labs/aB3r93yA

Practice Exercise Shopease

Workflow





Shopease Multi-Agent Customer Int...



shopease_customers



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	customer	customer_name	email	age	gender	location	signup_date	last_purchase	total_order	total_spent	avg_order	preferred_payment	loyalty_tier	churn_risk	
2	CUST0001	Samantha Davis	samantha.davis13	30	Male	El Paso, TX	1/21/2025	5/22/2025	5	137.25	29.98	Google Pay	0	Bronze	High
3	CUST0002	Brenda Gonzalez	brenda.gonzalez9	50	Non-binary	Dallas, TX	1/14/2022	6/11/2026	55	7633.04	154.78	PayPal	0	Gold	Low
4	CUST0003	Elizabeth Harris	elizabeth.harris22	44	Non-binary	Dallas, TX	3/30/2022	5/18/2026	16	6308.47	409.19	Google Pay	2	Gold	Low
5	CUST0004	Benjamin Flores	benjamin.flores56	24	Female	Atlanta, GA	8/16/2023	4/28/2026	30	10594.68	366.33	Apple Pay	2	Platinum	Low
6	CUST0005	Christopher White	christopher.white	39	Male	Nashville, TN	12/17/2022	3/23/2026	21	3985.01	203.83	Buy Now Pay Later	1	Gold	Medium
7	CUST0006	Debra Lee	debra.lee17@gmail	24	Male	Fort Worth, TX	7/3/2023	3/9/2025	41	4241.7	103.48	Buy Now Pay Later	7	Gold	High
8	CUST0007	Charles Taylor	charles.taylor27@	57	Non-binary	Dallas, TX	4/12/2025	2/8/2026	52	9771.67	163.52	Debit Card	3	Platinum	Medium
9	CUST0008	Elizabeth Carter	elizabeth.carter16	32	Female	Nashville, TN	5/11/2022	2/6/2026	60	15798.2	230.7	Credit Card	0	Platinum	Medium
10	CUST0009	Debra Wright	debra.wright44@y	40	Female	Philadelph	7/18/2024	8/12/2025	65	21128.73	324.31	Credit Card	4	Platinum	High

Shopease



<https://www.linkedin.com/pulse/multi-agents-orchestration-michael-lively-erawe/>

Test Cases

"Who are our highest churn risk customers?"

Customer Profile Agent

Segmented list with churn scores & loyalty tiers

"What's our payment failure rate this month?"

Payment & Transaction Agent

% failure breakdown by payment method

"How should we re-engage lapsed Gold tier users?"

Retention & Recommendation Agent

Prioritised win-back campaign actions

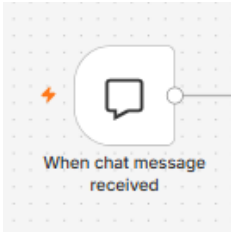
"Give me a full retention report"

All three agents

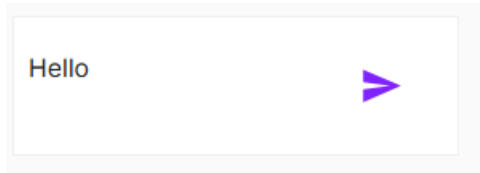
Synthesised multi-section report

Shopease Step-by-Step

To add this



Add the Chat Trigger



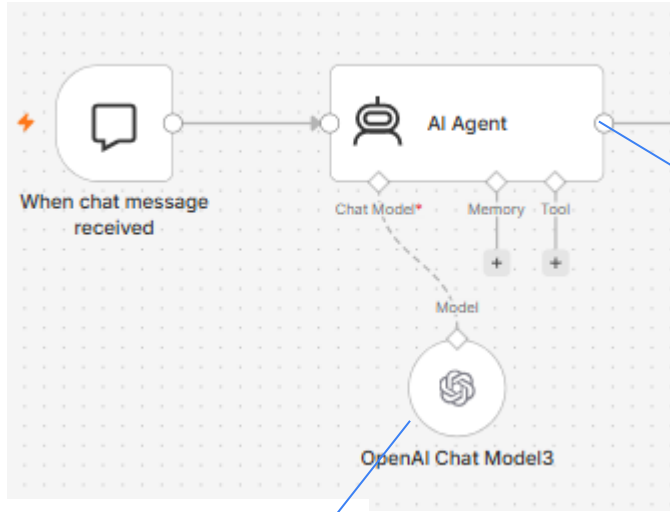
OUTPUT  

1 item

action	sessionId	chatInput
sendMessage	b21b017692df4b7ea645f9668b5c1155	Hello

Shopease Step-by-Step

System Message



To add this node



 AI Agent

 OpenAI

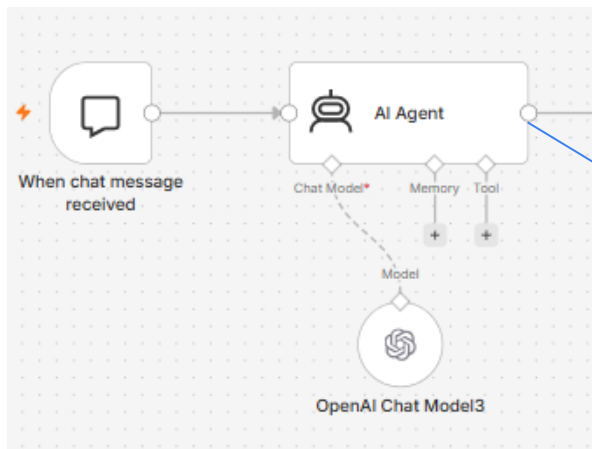
 OpenAI Chat Model



To add this node

Shopease Step-by-Step

System Message



Double Click the AI Agent Node

Change Name if Needed

AI Agent

INPUT

Parameters Settings

Execute step

Enable Fallback Model

Options

No properties

Add Option

System Message

Max Iterations

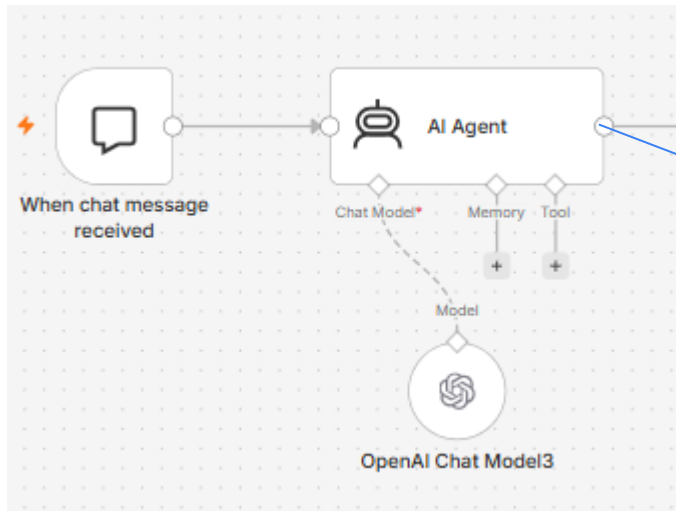
Return Intermediate Steps

Automatically Passthrough Binary Images

Chat Session: e9

Shopease Step-by-Step

System Message



You are an input pre-processor for a customer analytics system.

When you receive a user message, do the following:

1. Identify the intent: is it a WHO question (customer profile), a WHAT question (payment/transaction), a WHAT TO DO question (retention), or a COMBINED question?
2. Rewrite the query in a clear, specific, analytics-friendly format
3. If the query is too vague (e.g. "tell me something interesting"), expand it into a specific analytics question
4. Return **ONLY** the cleaned, rewritten query — nothing else

Example:

Input: "who is about to leave us?"

Output: "Identify all customers with High churn risk, their loyalty tier, last purchase date, and total spend"

Shopease Step-by-Step

You are an input pre-processor for a customer analytics system.

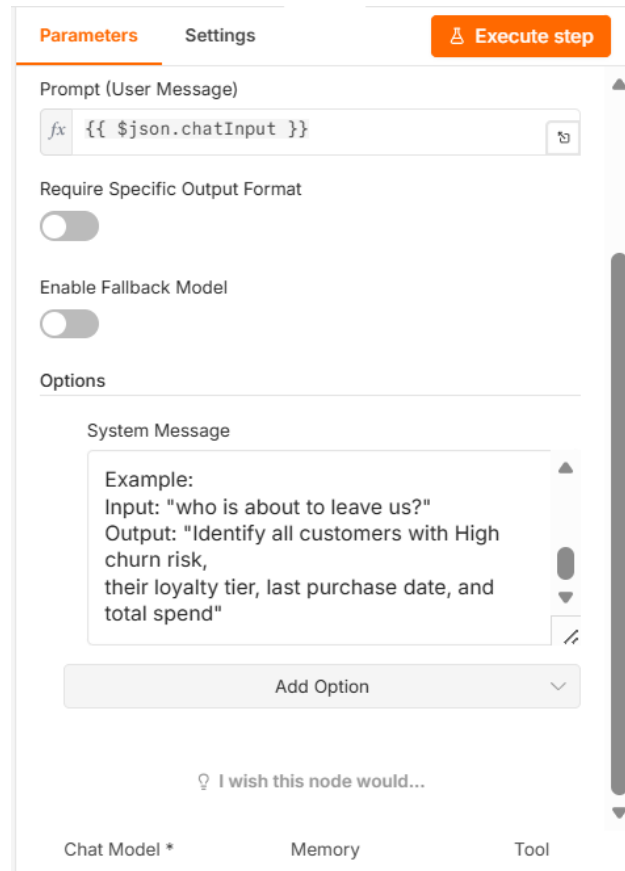
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Example:

Input: "who is about to leave us?"

Output: "Identify all customers with High churn risk, their loyalty tier, last purchase date, and total spend"



Parameters Settings **Execute step**

Prompt (User Message)

`fx {{ $json.chatInput }}`

Require Specific Output Format ☐

Enable Fallback Model ☐

Options

System Message

Example:
Input: "who is about to leave us?"
Output: "Identify all customers with High churn risk, their loyalty tier, last purchase date, and total spend"

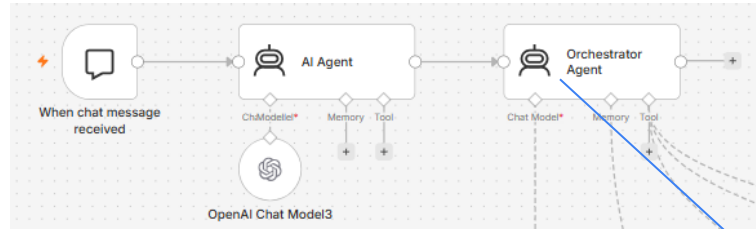
Add Option

I wish this node would...

Chat Model * Memory Tool

Shopease Step-by-Step

System Message



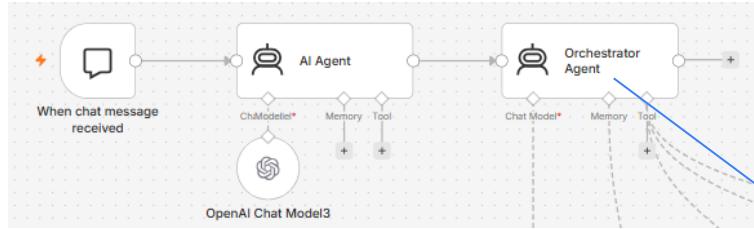
To add this node



 ai

 AI Agent

Shopease Step-by-Step



Double Click the Orchestrator Node

System Message

Change Name To Orchestrator Agent

AI Agent

INPUT

Parameters Settings

Execute step

Enable Fallback Model

Options

No properties

Add Option

System Message

Max Iterations

Return Intermediate Steps

Automatically Passthrough Binary Images

Chat Session: e9

Shopease Step-by-Step

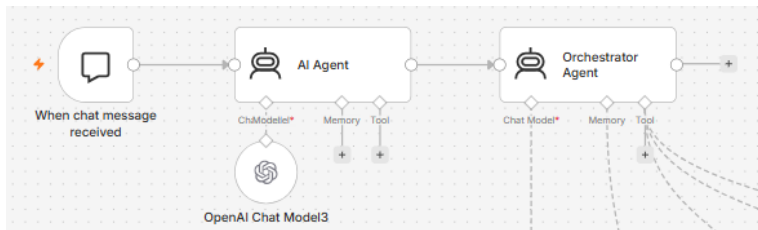
System Message

You are the Orchestrator Agent for the Shopease Customer Intelligence Hub.
Your job is to:

1. Understand the user's question or request.
2. Decide which specialist agent(s) should handle it:
 - Customer Profile Agent: questions about WHO customers are — churn risk, segments, demographics, loyalty tiers, lifetime value, behavioral patterns.
 - Payment & Transaction Agent: questions about WHAT happened transactionally — payment methods, failures, revenue, average order values, spend trends.
 - Retention & Recommendation Agent: questions about WHAT TO DO — win-back strategies, offers, loyalty upgrades, re-engagement campaigns, channel suggestions.
3. Route the request to the correct specialist tool(s).
4. Receive the specialist's output and synthesize a clean, clear, decision-friendly final response for the user.

Rules:

- Never analyze data yourself — always delegate to the right specialist.
- If a question spans multiple specialists (e.g., 'give me a full retention report'), call all relevant specialists and merge their outputs into one coherent response.
- Keep final responses structured, concise, and insight-driven.
- Maintain conversation context across turns using memory.





Shopease Step-by-Step

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System Message

Parameters Settings Execute step

Tip: Get a feel for agents with our quick [tutorial](#) or see an [example](#) of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

fx

{{ \$json.output }}

Require Specific Output Format

Enable Fallback Model

Options

System Message

You are the Orchestrator Agent for the Shopease Customer Intelligence Hub. Your job is to:

1. Understand the user's question or request.

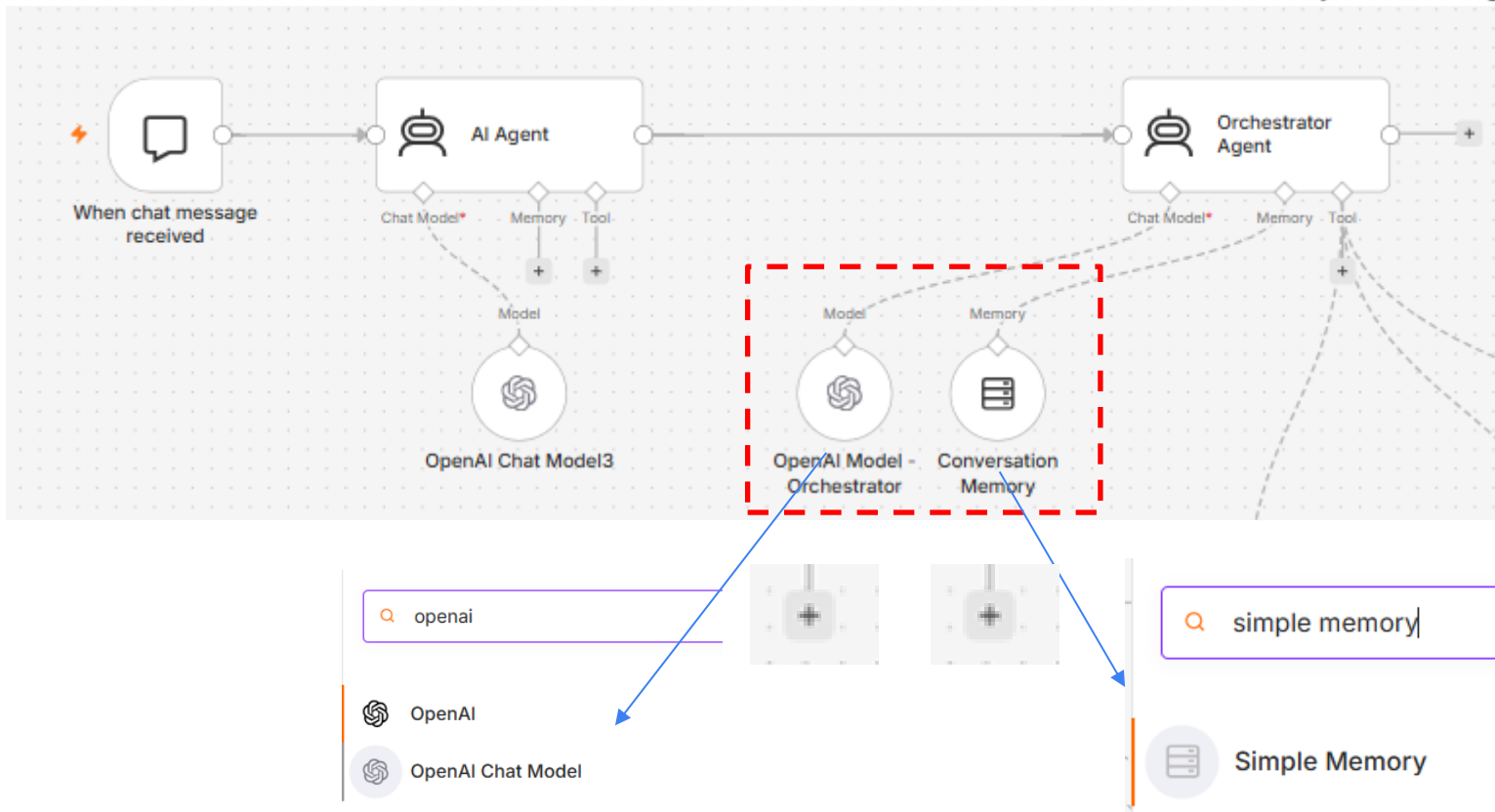
2. Decide which specialist agent(s) should

Chat Model * Memory Tool



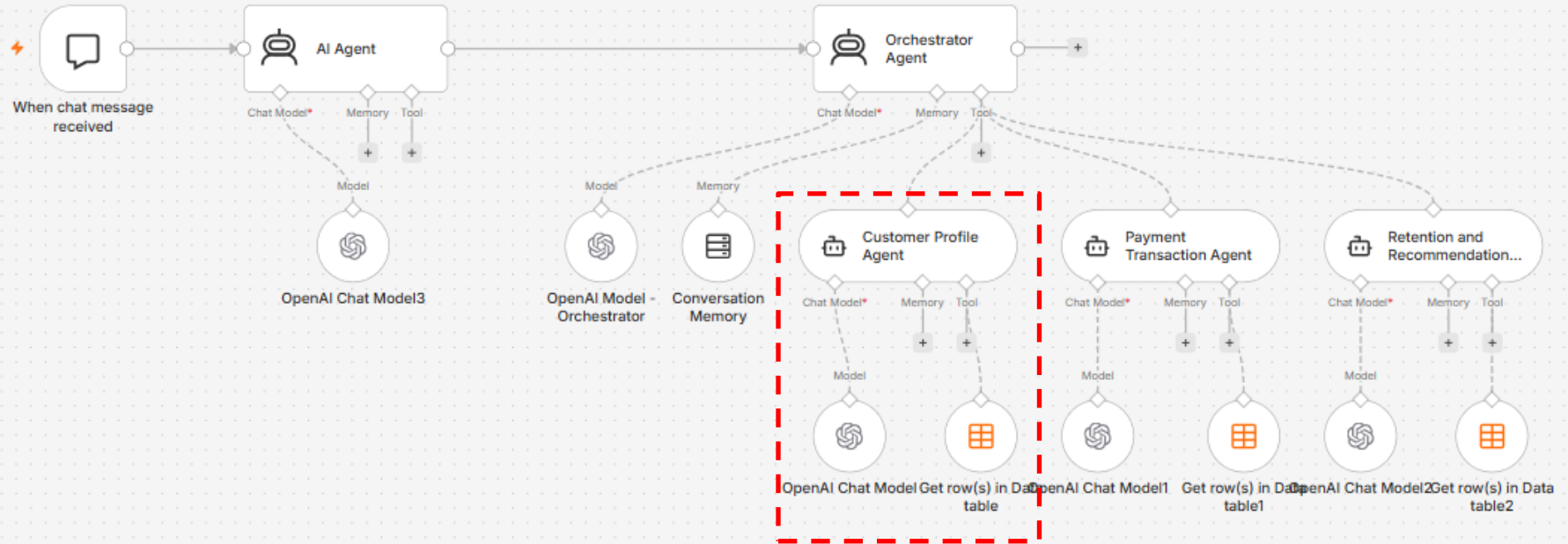
Shopease Step-by-Step

System Message



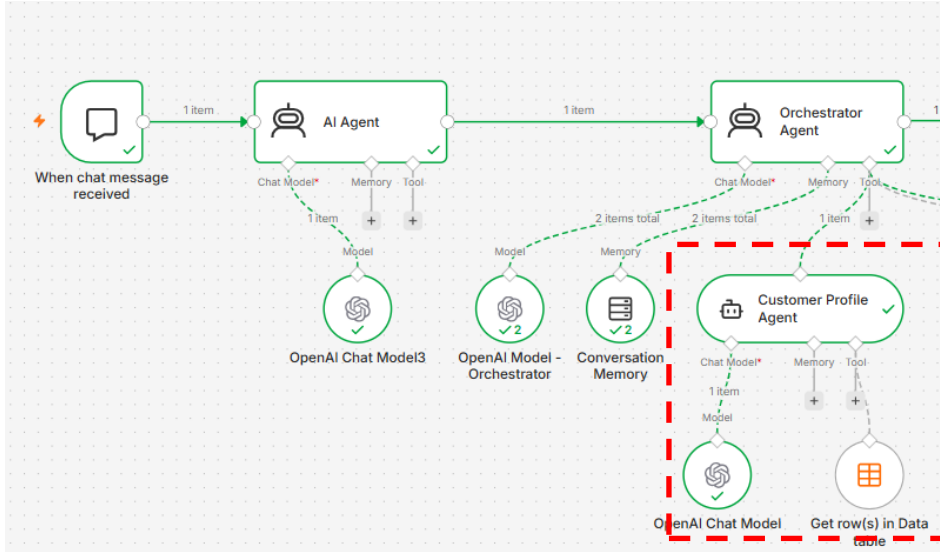
Shopease Step-by-Step

System Message



Shopease Step-by-Step

System Message

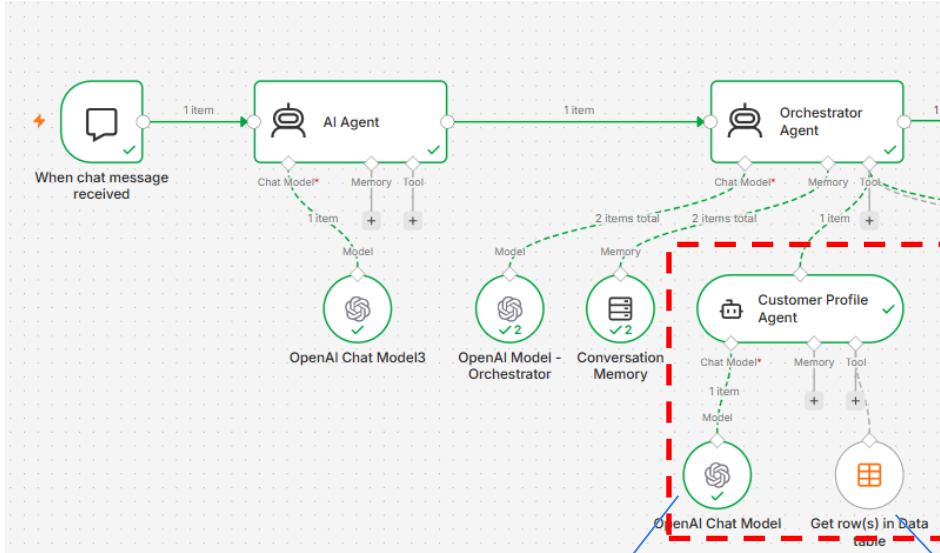


Connect to Tools

 **AI Agent**

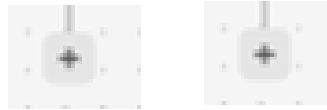
Shopease Step-by-Step

System Message



Search: openai

- OpenAI
- OpenAI Chat Model



Search: data table

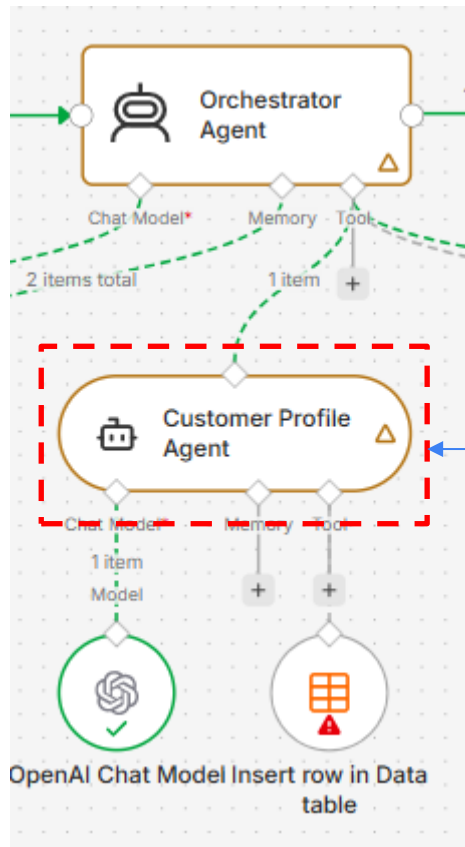
Data table Recommended
Permanently save data across workflow executions in a table

Data table Tool Recommended



Shopease Step-by-Step

System Message



Change the name and
add the description.

Double Click

Change Name To  Customer Profile Agent

Description

AI Agent that can call other tools

Prompt (User Message)

e.g. Hello, how can you help me?



Description

Use this tool for questions about WHO customers are — churn risk, loyalty tiers, demographics, lifetime value, recency, and segmentation

Prompt (User Message)

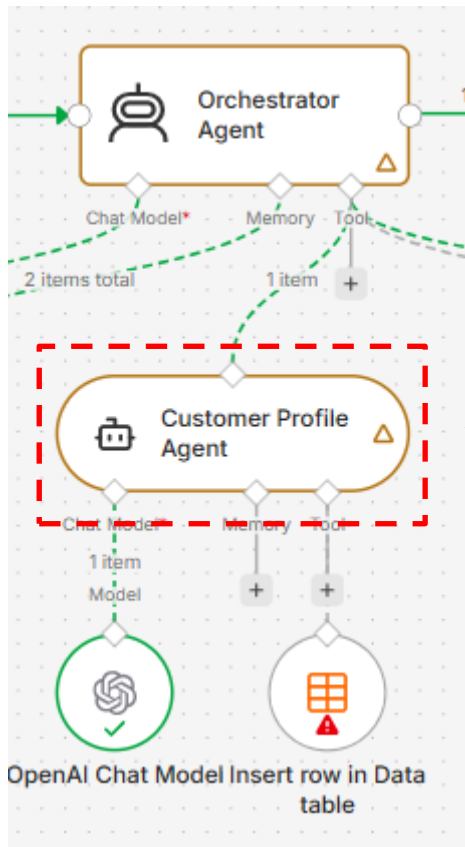
`fx {{ $json.output }}`



Shopease Step-by-Step

System Message

Add System Message



Enable Fallback Model



Options

No properties

Add Option

💡 I wish this node would...

Chat Model *

Memory

Tool

System Message

Max Iterations

Return Intermediate Steps

Automatically Passthrough Binary Images

Automatically Passthrough Binary PDFs

Tracing Metadata

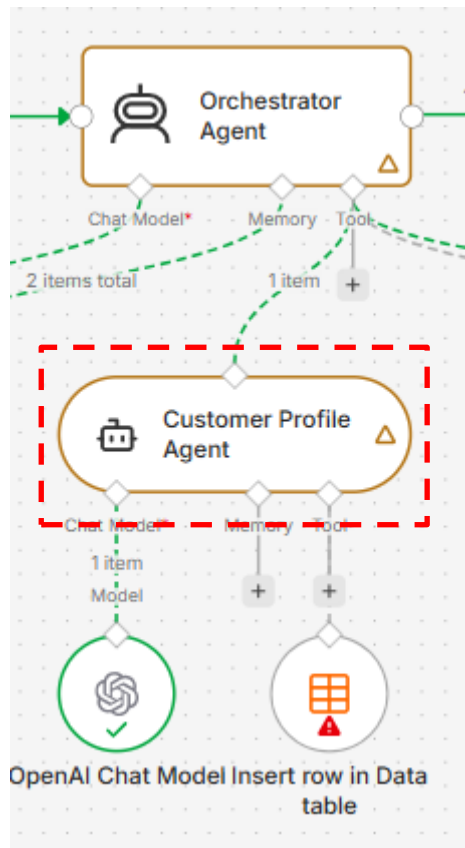
Auto-save highlighted data

Enable Streaming

Shopease Step-by-Step

System Message

Add System Message



Parameters

Settings

 Execute step

System Message

You are the Customer Profile Agent for Shopease. Only answer WHO questions — churn risk, loyalty tiers, demographics, LTV, recency, signup trends. Use only data from the data table tool. Provide insights in clear bullet points with numbers and percentages. Do not handle payment or recommendation questions.

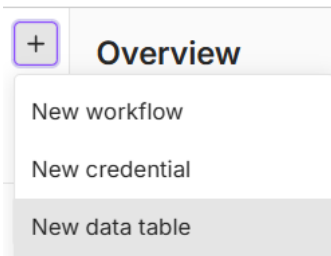
Add Option

 I wish this node would...



Set Up the Table

Hit the plus button and select
New data table



Give it the name **Shopeasepayments**

Create new data table



Double click the table and make sure
Shopeasepayments is selected.

Data table name *

Shopeasepayments

☐ From scratch

☒ Import CSV

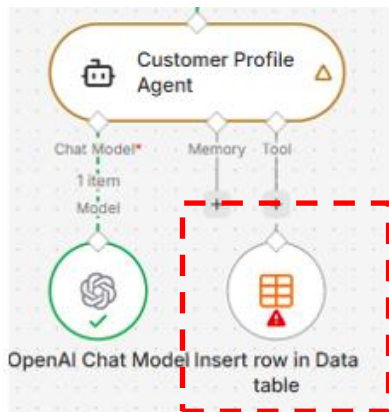
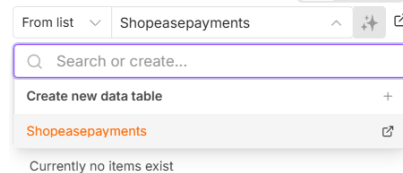


shopease_customers.csv

☒ My CSV file contains a header row

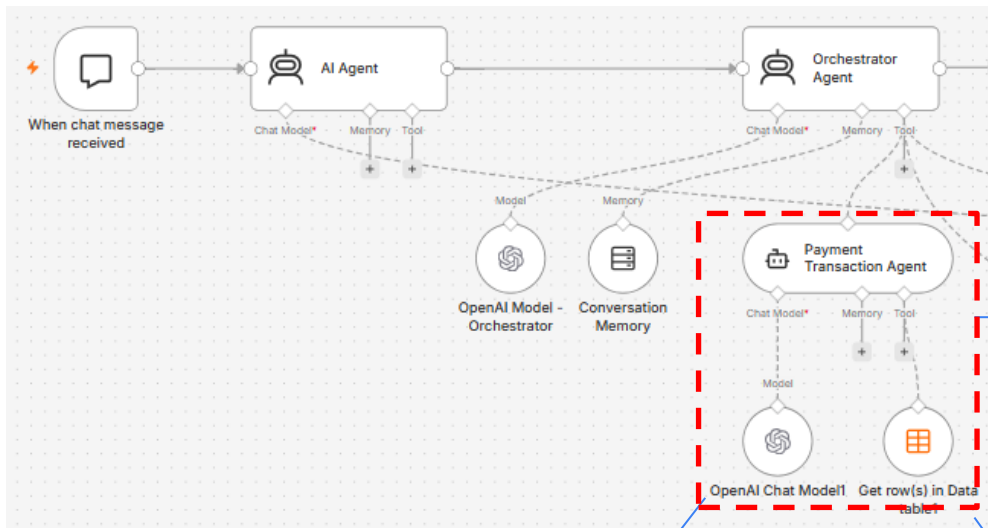
Cancel

Create



Shopease Step-by-Step

System Message



Connect to Tools

ai

AI Agent

openai

OpenAI

OpenAI Chat Model

data table

Data table Recommended

Permanently save data across workflow executions in a table

Data table Tool Recommended



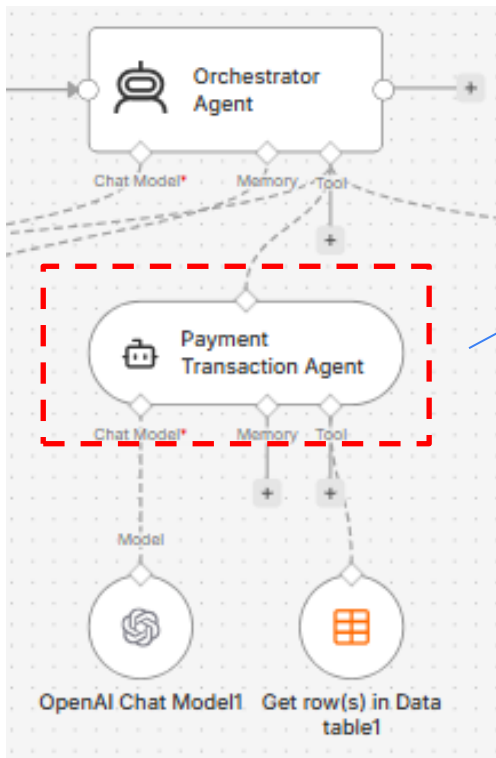
Shopease Step-by-Step

System Message

Change Name To



Payment Transaction Agent



Description

AI Agent that can call other tools

Prompt (User Message)

e.g. Hello, how can you help me?

Description

Use this tool for questions about WHAT happened transactionally — payment methods, failures, revenue, average order values, and spend patterns

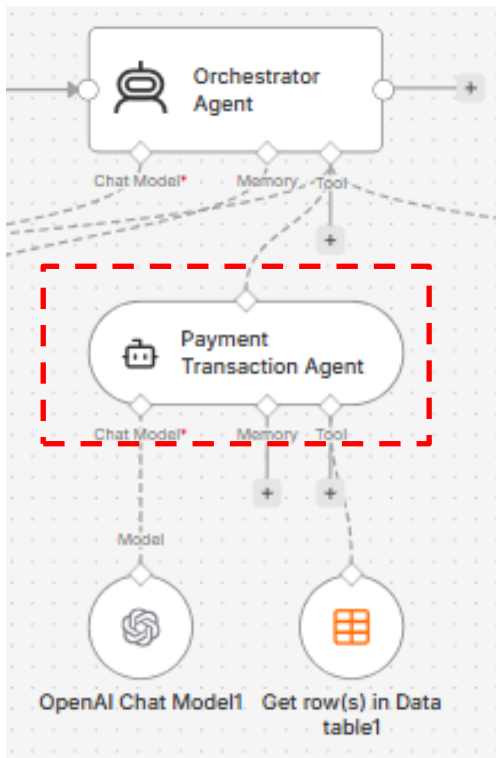
Prompt (User Message)

`fx {{ $json.output }}`



Shopease Step-by-Step

System Message



Enable Fallback Model



Options

No properties

Add Option

💡 I wish this node would...

Chat Model *

Memory

Tool

System Message

Max Iterations

Return Intermediate Steps

Automatically Passthrough Binary Images

Automatically Passthrough Binary PDFs

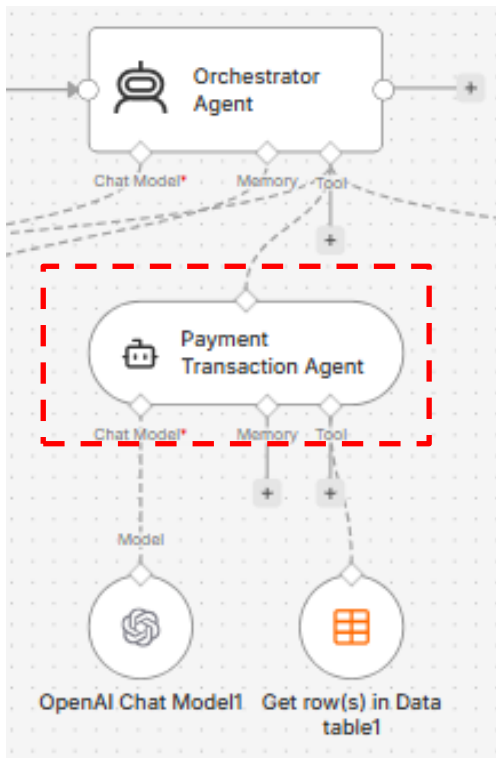
Tracing Metadata

Auto-save highlighted data

Enable Streaming



Shopease Step-by-Step



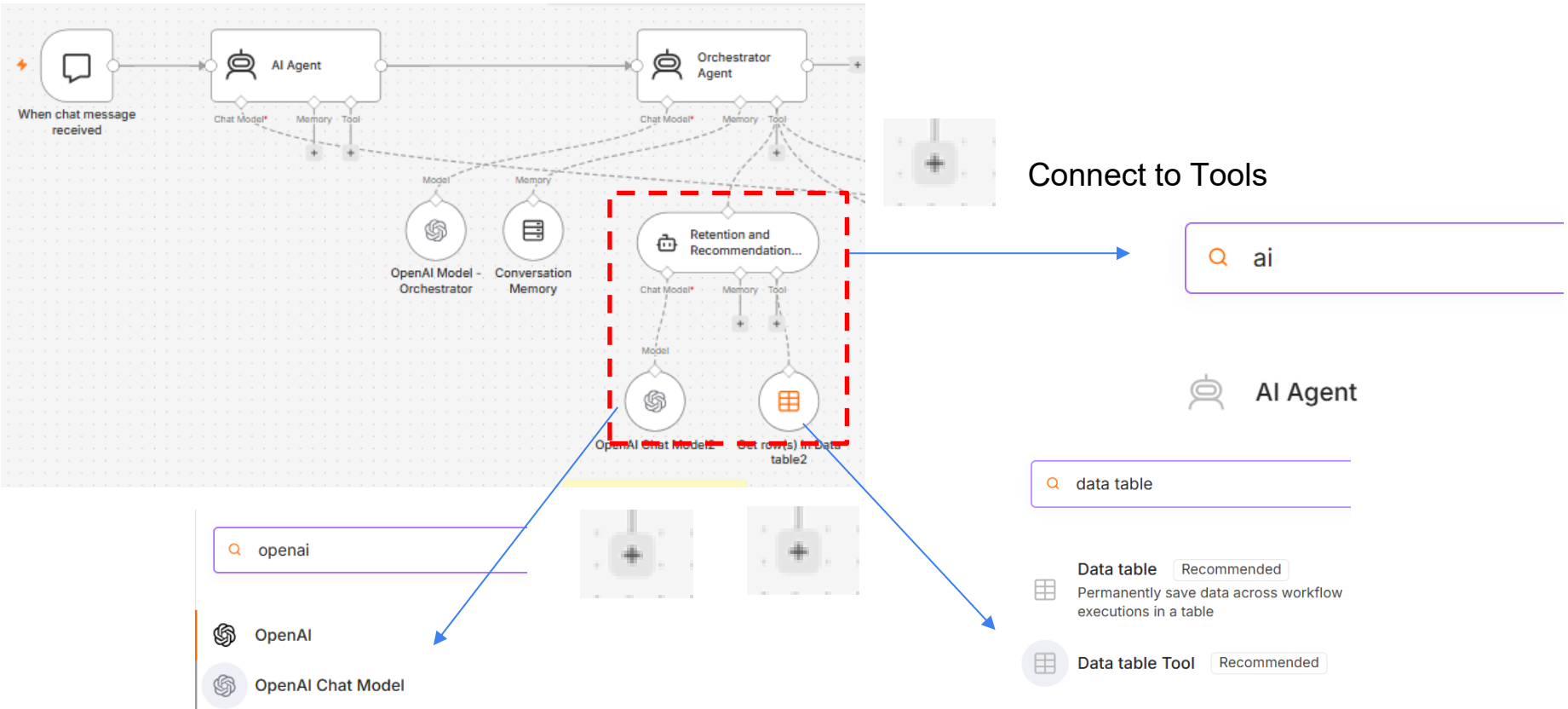
System Message

Parameters Settings **Execute step**

Options

System Message

You are the Payment & Transaction Agent for Shopease. Only answer WHAT happened questions — payment method distribution, failure rates, revenue, average order value, spend breakdown by tier or location. Use only data from the data table tool. Provide insights in bullet points with numbers. Do not handle customer profiling or recommendation questions.



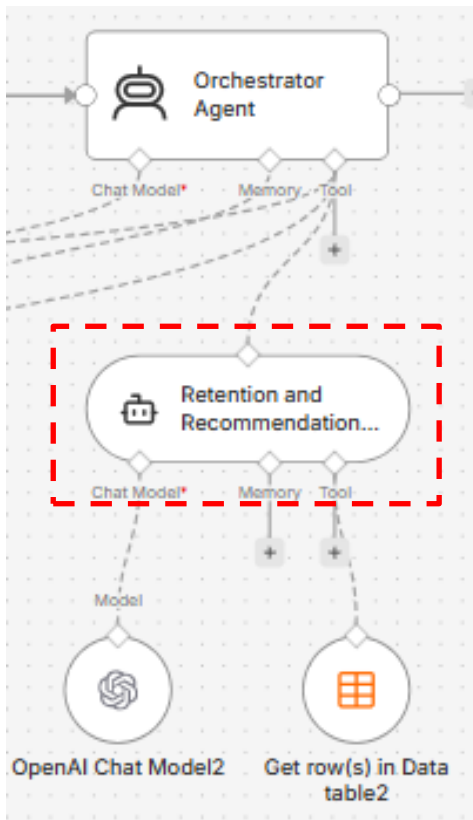
Shopease Step-by-Step

System Message

Change Name To



Retention and Recommendation Agent



Description

AI Agent that can call other tools

Prompt (User Message)

e.g. Hello, how can you help me?

Description

Use this tool for questions about WHAT TO DO — win-back offers, loyalty upgrade nudges, re-engagement campaigns, payment friction fixes, and retention strategies

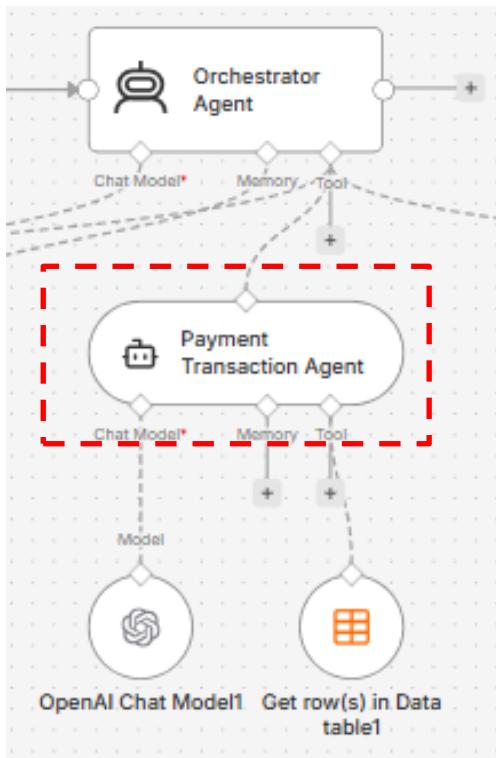
Prompt (User Message)

`fx {{ $json.output }}`



Shopease Step-by-Step

System Message



Enable Fallback Model



Options

No properties

Add Option

💡 I wish this node would...

Chat Model *

Memory

Tool

System Message

Max Iterations

Return Intermediate Steps

Automatically Passthrough Binary Images

Automatically Passthrough Binary PDFs

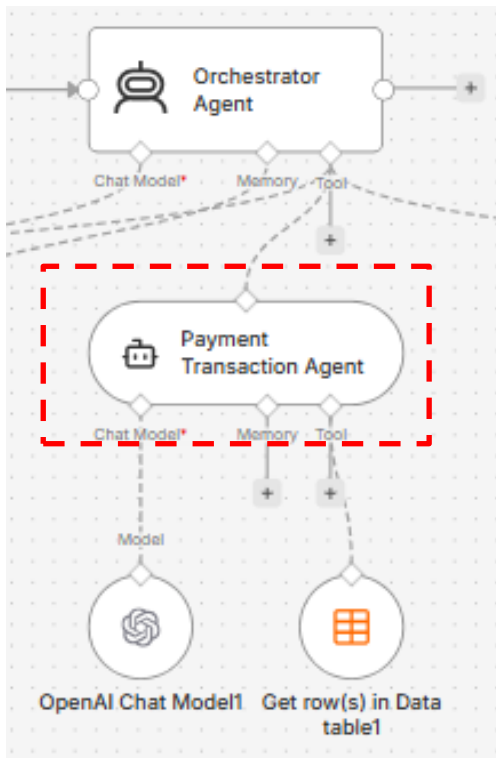
Tracing Metadata

Auto-save highlighted data

Enable Streaming

Shopease Step-by-Step

System Message



Parameters

Settings

 Execute step

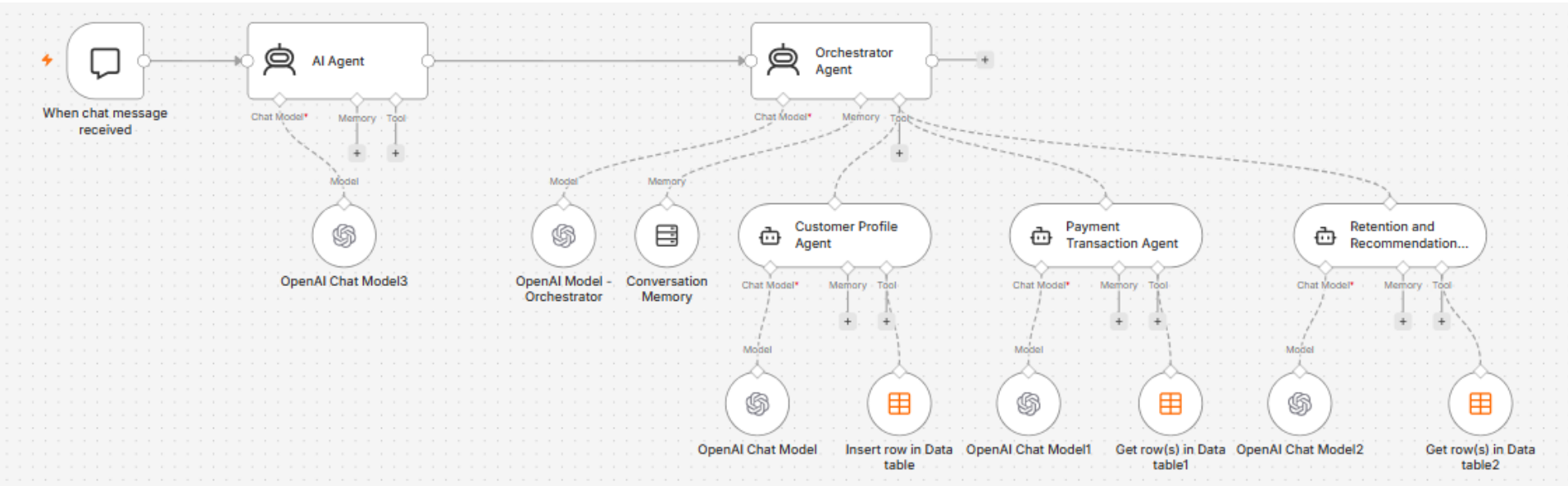
Options

System Message

You are the Retention & Recommendation Agent for Shopease. Only answer WHAT TO DO questions — win-back campaigns, loyalty upgrade nudges, payment friction fixes, re-engagement timing, and segment-specific offers. Base all recommendations strictly on data from the data table tool. Format output as prioritised action lists. Do not perform customer profiling or payment analysis.

Shopease Step-by-Step

System Message



Test Cases

"Who are our highest churn risk customers?"

Customer Profile Agent

Segmented list with churn scores & loyalty tiers

"What's our payment failure rate this month?"

Payment & Transaction Agent

% failure breakdown by payment method

"How should we re-engage lapsed Gold tier users?"

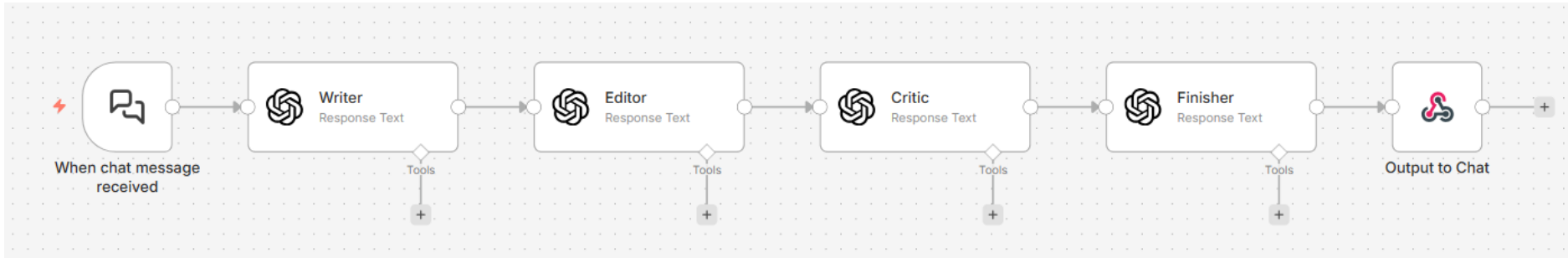
Retention & Recommendation Agent

Prioritised win-back campaign actions

"Give me a full retention report"

All three agents

Synthesised multi-section report



create four prompts for a story creator, the story should be 4 pages in length and use a writer editor critic and finisher system prompts also give the user prompts for each n8n message to mode node, the chat input will be used to create the story

<https://quantumai-lively.app.n8n.cloud/workflow/KAPd1NOPOJh0Bqfn>

<https://www.linkedin.com/pulse/building-agents-n8n-basics-course-michael-lively-to0rc/>

Pushover

Android

iPhone & iPad

Desktop

Integrations

Pricing

Teams

API

Blog

Help

Login or Signup

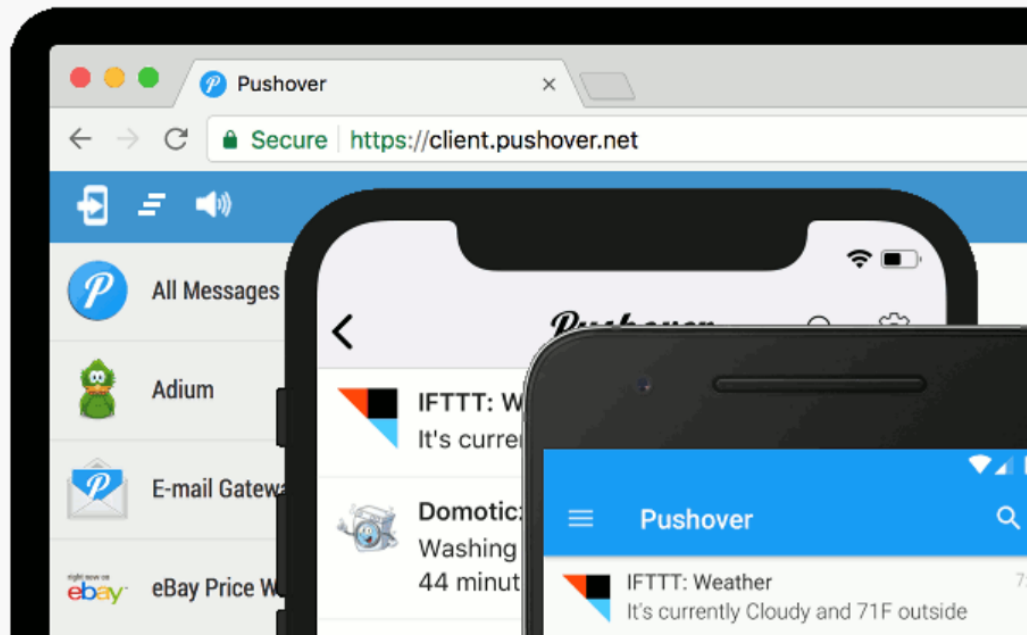
Latest Pushover News: API usage limit changes coming May 1st

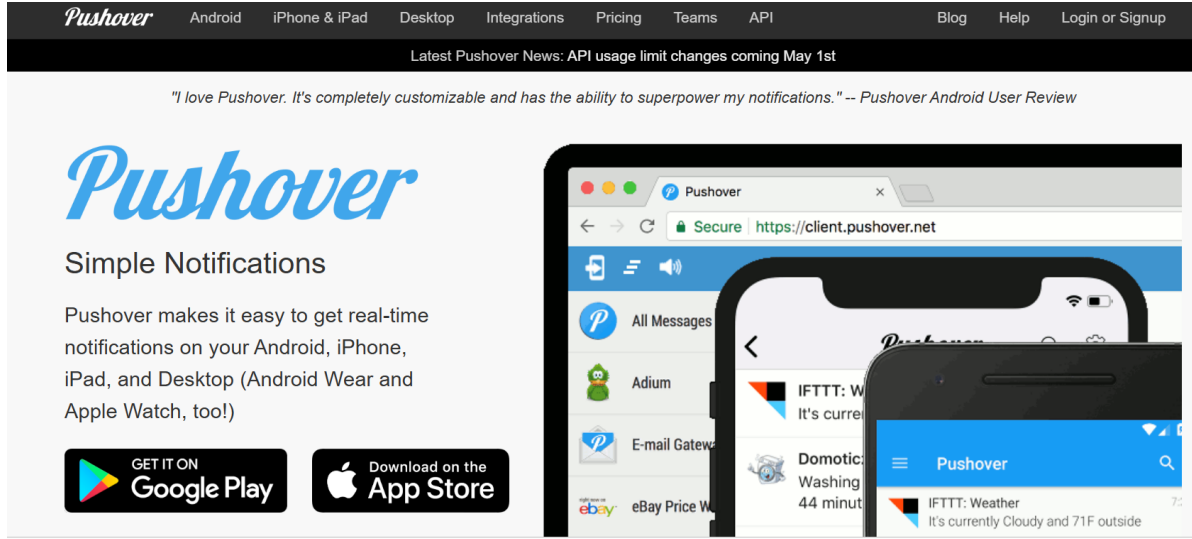
"I love Pushover. It's completely customizable and has the ability to superpower my notifications." -- Pushover Android User Review

Pushover

Simple Notifications

Pushover makes it easy to get real-time notifications on your Android, iPhone, iPad, and Desktop (Android Wear and Apple Watch, too!)

GET IT ON
Google PlayDownload on the
App Store



The screenshot shows the Pushover website. At the top is a navigation bar with links: Pushover, Android, iPhone & iPad, Desktop, Integrations, Pricing, Teams, API, Blog, Help, and Login or Signup. Below the navigation bar is a black banner with white text: "Latest Pushover News: API usage limit changes coming May 1st". Underneath is a quote: "I love Pushover. It's completely customizable and has the ability to superpower my notifications." -- Pushover Android User Review. The main content area features the Pushover logo in blue script, followed by the text "Simple Notifications". Below this, it says "Pushover makes it easy to get real-time notifications on your Android, iPhone, iPad, and Desktop (Android Wear and Apple Watch, too!)". At the bottom of the main content area are two buttons: "GET IT ON Google Play" and "Download on the App Store". To the right of the text is a large image showing a desktop browser window displaying the Pushover website, with a smartphone and a tablet overlaid on top, showing the Pushover app interface with various notifications like "IFTTT: Weather", "Domoticz: Washing", and "eBay Price Watch".

Create an account and you will see an API key!

You will need two tokens.

<https://pushover.net/>

URL:

optional

Send Notification

Your Devices [\(Add Phone, Tablet, or Desktop\)](#) [\(View Your Licenses\)](#)

Name	Status	Client	Last Seen
 toby	Enabled	Pushover for iOS 4.3.2	4 days ago
 toby17	Enabled	Pushover for iOS 4.3.3	about 17 hours ago

Your E-Mail Aliases [\(Create an E-Mail Alias\)](#)

Address	Settings
 hqx9m1d7h4@pomail.net	Deliver to all of your devices (default)

Your Delivery Groups [\(Create a Group\)](#)

No delivery groups created yet.

Your Custom Sounds [\(Upload a Sound\)](#)

No custom sounds uploaded yet.

Your Applications [\(Create an Application/API Token\)](#)

Name	Description	Messages Sent / Allowed
 deal		76 / 10,000

Your Invoices

Invoice ID	Description / Amount	Paid On
 241130202654	License for ed.donner@gmail.com - u1oju2di9e3tff65mbaq9ffnen5ob (iOS) (Sale price), \$2.49	2024-11-30

Give you application a name and get your application token (will start with an a)



Two tokens on starts with a u (user) and a (application).

<https://pushover.net/>

Install on your phone
the pushover app



URL:

optional

Send Notification

Your Devices [\(Add Phone, Tablet, or Desktop\)](#) [\(View Your Licenses\)](#)

Name	Status	Client	Last Seen
 toby	Enabled	Pushover for iOS 4.3.2	4 days ago
 toby17	Enabled	Pushover for iOS 4.3.3	about 17 hours ago

Your E-Mail Aliases [\(Create an E-Mail Alias\)](#)

Address	Settings
 hqx9m1d7h4@pomail.net	Deliver to all of your devices (default)

Your Delivery Groups [\(Create a Group\)](#)

No delivery groups created yet.

Your Custom Sounds [\(Upload a Sound\)](#)

No custom sounds uploaded yet.

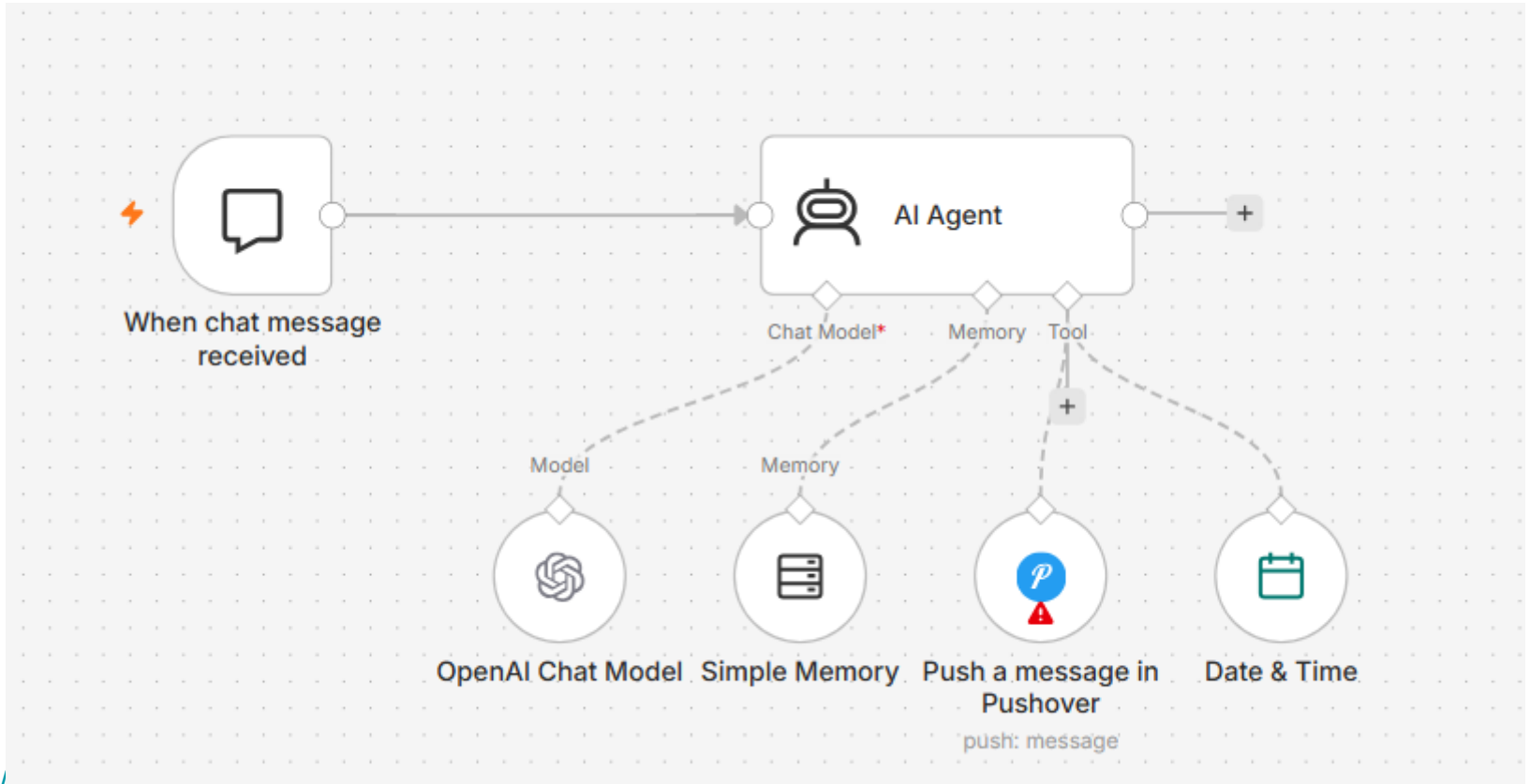
Your Applications [\(Create an Application/API Token\)](#)

Name	Description	Messages Sent / Allowed
 deal		76 / 10,000

Your Invoices

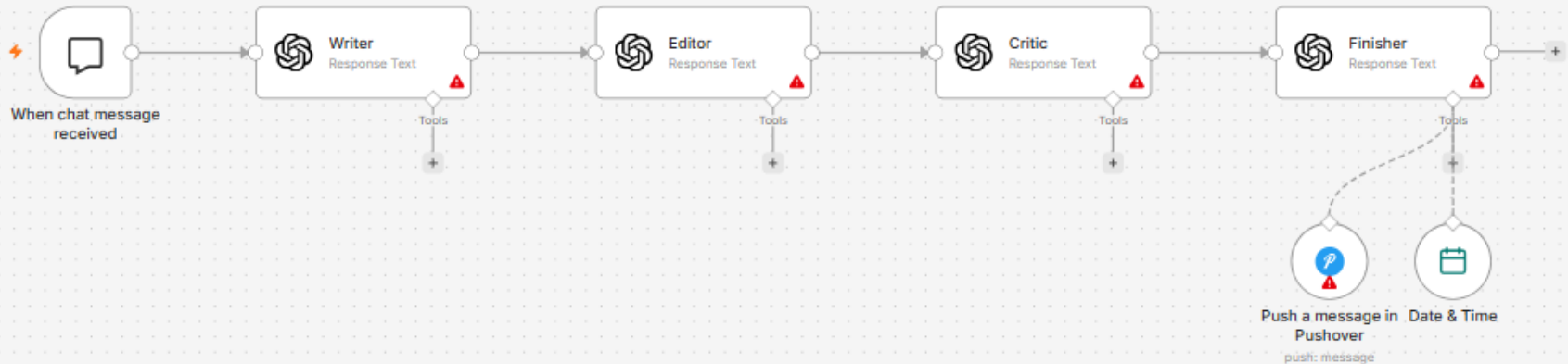
Invoice ID	Description / Amount	Paid On
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<https://pushover.net/>



<https://pushover.net/>

Push Notification Multi-Agent Story Creator



<https://pushover.net/>

Summary

Learning Objectives

Copilot Help Desk

What is wrong?

Why this error cascades?.

Shopease Hands-On Step-by-Step

Slides and Prompts will be provided

Mobile Push (Walk Through – How you Do It!)

Adding mobile push

Adding mobile push to the simple story maker

Session Summary



Power Ahead!



APPENDIX